

Definition of variables and format used in the GLOF database. [STR a string; INT an integer; * cases where only inputs from a predescribed list are possible; NA if unknown; + if exact number is unavailable; Y - Yes; N - No].

Variable	Format	Description and unit
GF_ID	INT	Unique identifier for each individual event, starting at 1.
Year_approx	STR	Year of occurrence; given approximately if GLOF has been identified from imagery with no definite account.
Year_exact	INT	Exact year of occurrence.
Month	INT	Month of occurrence.
Day	INT	Day of occurrence.
Lake_name	STR	Local name of the lake, if available.
Glacier_name	STR	Local name of the glacier associated with the lake, if available.
GL_ID	STR	Unique identifier for the lake.
G_ID	STR	Unique identifier for the glacier (GLIMS format) associated to the lake.
LakeDB_ID	INT	Identifier in which lakes database the lake has been mapped. 2- Wang2020 (1990) ; 3- Wang2020 (2018) ; 4- Chen2021
Lat_lake	INT	Decimal degrees latitude of lake [°, WGS 84].
Lon_lake	INT	Decimal degrees longitude of lake [°, WGS 84].
Elev_lake	INT	Elevation of lake [m a.s.l.].
Lat_impact	INT	Decimal degrees latitude of lowest known impact of the GLOF [°, WGS 84].
Lon_impact	INT	Decimal degrees longitude of lowest known impact of the GLOF [°, WGS 84].
Elev_impact	INT	Elevation of lowest known impact of the GLOF [m a.s.l.].
Impact_type	STR*	Quality of impact record; ‘Observation’ refers to lowest observed high flow or damages; ‘Deposit’ refers to lowest visible deposit of sediments (from satellite imagery). Impacts are therefore conservative and are likely always further downstream than recorded.
Lake_type	STR*	Type of lake (e.g., moraine dammed, ice dammed, bedrock etc.).
Transboundary	STR*	Denotes if impact is potentially transboundary.
Repeat	STR*	GLOF that occurred again in past or future.
Region_RGI	STR	Region ID as used in the RGI 6.0 (Pfeffer, 2017).
Region_HiMAP	INT	Region ID as used in the HiMAP report (Bolch et al., 2019).
Country	STR	Current national borders (2022) the source lake lies in.
Province	STR	1 st level province name.
River Basin	STR	River basin the lake is located in.
Driver_lake	STR*	Driver that caused the lake to form.
Driver_GLOF	STR*	Driver that caused the GLOF to occur.
Mechanism	STR*	Mechanism involved in lake breach or drainage.
Area	INT	Lake area [m ²].

Variable	Format	Description and unit
Volume	INT	Volume of the drained lake water [m ³].
Discharge_water	INT	Measured or estimated water discharge downstream [m ³ s ⁻¹].
Discharge_solid	INT	Measured or estimated debris flow discharge downstream [m ³ s ⁻¹].
Impact	STR	Narrative description of downstream impacts.
Lives_total	INT	Total lives lost.
Lives_male	INT	Total male lives lost.
Lives_female	INT	Total female lives lost.
Lives_disabilities	INT	Total lives of people with disabilities lost.
Injured_total	INT	Total injured.
Injured_male	INT	Total male injured.
Injured_female	INT	Total female injured.
Injured_disabilities	INT	Total people with disabilities injured.
Displaced_total	INT	Total people displaced.
Displaced_male	INT	Total male displaced.
Displaced_female	INT	Total female displaced.
Displaced_disabilities	INT	Total people with disabilities displaced.
Livestock	INT	Livestock lost.
Residential_destroyed	INT	Number of residential houses destroyed.
Commercial_destroyed	INT	Number of commercial houses destroyed.
Residential_damaged	INT	Number of residential houses damaged.
Commercial_damaged	INT	Number of commercial houses damaged.
Infra	STR	Other destroyed infrastructure.
Agricultural	INT	Area of farmland destroyed [m ²].
Hydropower	INT	Installed hydropower capacity destroyed [MW] .
Econ_damage	INT	Total economic damage [USD].
Sat_evidence	STR	Image IDs used as satellite imagery evidence.
Ref_scientific	STR	Citation of scientific source.
Ref_scientific_full	STR	Full references to the scientific sources.
Ref_other	STR	Description of other sources.
Remarks	STR	Any other remarks.